

EE708 Assignment 4

1. Gini Index

Total samples = 200, Positive = 120, Negative = 80

(a) Gini before split

$$\begin{aligned}p_+ &= \frac{120}{200} = 0.6, & p_- &= \frac{80}{200} = 0.4 \\Gini &= 1 - (p_+^2 + p_-^2) = 1 - (0.6^2 + 0.4^2) \\&= 1 - (0.36 + 0.16) = 0.48\end{aligned}$$

(b) After split

Left node: (50+, 10-), total = 60

$$\begin{aligned}p_+ &= \frac{50}{60}, & p_- &= \frac{10}{60} \\Gini_L &= 1 - \left(\left(\frac{50}{60} \right)^2 + \left(\frac{10}{60} \right)^2 \right) = 1 - \left(\frac{2500}{3600} + \frac{100}{3600} \right) = 1 - 0.7222 = 0.2778\end{aligned}$$

Right node: (70+, 70-), total = 140

$$\begin{aligned}p_+ &= 0.5, & p_- &= 0.5 \\Gini_R &= 1 - (0.5^2 + 0.5^2) = 0.5\end{aligned}$$

Weighted Gini:

$$\begin{aligned}Gini_{split} &= \frac{60}{200}(0.2778) + \frac{140}{200}(0.5) \\&= 0.3(0.2778) + 0.7(0.5) = 0.0833 + 0.35 = 0.4333\end{aligned}$$

$$0.4333 < 0.48 \Rightarrow \text{Split improves purity}$$

2. Regression Tree using SSE

Given data:

$$(1, 5, 10), (2, 6, 12), (3, 8, 15), (4, 10, 18), (5, 12, 21), (6, 15, 25), (7, 18, 28), (8, 20, 30)$$

We test split at $x_1 = 4.5$

Left: $x_1 \leq 4$

$$y = \{10, 12, 15, 18\}$$

Mean:

$$\bar{y}_L = \frac{10 + 12 + 15 + 18}{4} = 13.75$$

$$SSE_L = \sum (y_i - 13.75)^2 = 14.06 + 3.06 + 1.56 + 18.06 = 36.74$$

Right: $x_1 > 4$

$$y = \{21, 25, 28, 30\}$$

Mean:

$$\bar{y}_R = 26$$

$$SSE_R = 25 + 1 + 4 + 16 = 46$$

$$Total\ SSE = 36.74 + 46 = 82.74$$

Best split: $x_1 = 4.5$ (minimum SSE)

3. Vector Quantization

Centroids:

$$C_1 = (2, 3), \quad C_2 = (5, 8), \quad C_3 = (9, 4)$$

$$(1, 2) \rightarrow C_1, \quad (3, 4) \rightarrow C_1$$

$$(6, 7), (5, 5) \rightarrow C_2$$

$$(8, 3) \rightarrow C_3$$

(a) New centroids

$$C'_1 = \left(\frac{1+3}{2}, \frac{2+4}{2} \right) = (2, 3)$$

$$C'_2 = \left(\frac{6+5}{2}, \frac{7+5}{2} \right) = (5.5, 6)$$

$$C'_3 = (8, 3)$$

(b) Distortion

Old distortion:

$$(1, 2) \rightarrow 2, \quad (3, 4) \rightarrow 2, \quad (6, 7) \rightarrow 2, \quad (5, 5) \rightarrow 9, \quad (8, 3) \rightarrow 2$$
$$= 17$$

New distortion:

$$= 2 + 2 + 1.25 + 1.25 + 0 = 6.5$$

It implies distortion decreases

4. Derivation of EM Updates

We maximize:

$$\mathcal{Q} = \sum_{n=1}^N \sum_{k=1}^K \gamma(z_{nk}) (\ln \pi_k + \ln \mathcal{N}(x_n | \mu_k, \Sigma_k))$$

Expand Gaussian:

$$\ln \mathcal{N}(x_n | \mu_k, \Sigma_k) = -\frac{1}{2} \ln |\Sigma_k| - \frac{1}{2} (x_n - \mu_k)^T \Sigma_k^{-1} (x_n - \mu_k) - \frac{D}{2} \ln(2\pi)$$

Ignore constants:

$$\mathcal{Q} = \sum_{k=1}^K \sum_{n=1}^N \gamma(z_{nk}) \left[\ln \pi_k - \frac{1}{2} \ln |\Sigma_k| - \frac{1}{2} (x_n - \mu_k)^T \Sigma_k^{-1} (x_n - \mu_k) \right]$$

Define:

$$N_k = \sum_{n=1}^N \gamma(z_{nk})$$

(a) Derivation of π_k

We maximize:

$$\sum_k N_k \ln \pi_k \quad \text{subject to} \quad \sum_k \pi_k = 1$$

Use Lagrange multiplier λ :

$$\mathcal{L} = \sum_k N_k \ln \pi_k + \lambda \left(\sum_k \pi_k - 1 \right)$$

Differentiate:

$$\frac{\partial \mathcal{L}}{\partial \pi_k} = \frac{N_k}{\pi_k} + \lambda = 0$$

$$\Rightarrow \pi_k = -\frac{N_k}{\lambda}$$

Sum over k :

$$\sum_k \pi_k = -\frac{1}{\lambda} \sum_k N_k = 1$$

$$\sum_k N_k = N \Rightarrow -\frac{N}{\lambda} = 1 \Rightarrow \lambda = -N$$

Thus:

$$\boxed{\pi_k = \frac{N_k}{N}}$$

(b) Derivation of Σ_k

Focus on:

$$\mathcal{Q}_k = -\frac{1}{2} \sum_n \gamma(z_{nk}) [\ln |\Sigma_k| + (x_n - \mu_k)^T \Sigma_k^{-1} (x_n - \mu_k)]$$

Take derivative w.r.t Σ_k^{-1} using identity:

$$\frac{\partial}{\partial A^{-1}} \ln |A| = A$$

$$\frac{\partial}{\partial \Sigma_k^{-1}} (x - \mu)^T \Sigma_k^{-1} (x - \mu) = (x - \mu)(x - \mu)^T$$

Thus:

$$\frac{\partial \mathcal{Q}_k}{\partial \Sigma_k^{-1}} = \frac{1}{2} \sum_n \gamma(z_{nk}) [\Sigma_k - (x_n - \mu_k)(x_n - \mu_k)^T]$$

Set to zero:

$$\sum_n \gamma(z_{nk}) \Sigma_k = \sum_n \gamma(z_{nk}) (x_n - \mu_k)(x_n - \mu_k)^T$$

$$\Sigma_k \sum_n \gamma(z_{nk}) = \sum_n \gamma(z_{nk}) (x_n - \mu_k)(x_n - \mu_k)^T$$

$$\boxed{\Sigma_k = \frac{1}{N_k} \sum_n \gamma(z_{nk}) (x_n - \mu_k)(x_n - \mu_k)^T}$$

5. Conditional Density is a Mixture

$$p(x) = \sum_k \pi_k p(x|k)$$

Partition:

$$x = (x_a, x_b)$$

Joint:

$$p(x_a, x_b) = \sum_k \pi_k p(x_a, x_b|k)$$

$$p(x_a, x_b|k) = p(x_b|x_a, k)p(x_a|k)$$

$$p(x_a, x_b) = \sum_k \pi_k p(x_a|k) p(x_b|x_a, k)$$

Marginal:

$$p(x_a) = \sum_j \pi_j p(x_a|j)$$

Conditional:

$$\begin{aligned} p(x_b|x_a) &= \frac{p(x_a, x_b)}{p(x_a)} \\ &= \frac{\sum_k \pi_k p(x_a|k) p(x_b|x_a, k)}{\sum_j \pi_j p(x_a|j)} \\ &= \sum_k \left(\frac{\pi_k p(x_a|k)}{\sum_j \pi_j p(x_a|j)} \right) p(x_b|x_a, k) \\ \tilde{\pi}_k(x_a) &= \frac{\pi_k p(x_a|k)}{\sum_j \pi_j p(x_a|j)} \end{aligned}$$

Thus:

$$p(x_b|x_a) = \sum_k \tilde{\pi}_k(x_a) p(x_b|x_a, k)$$

Hence proven it is a mixture.

6. Full Likelihood and MLE

(a) Likelihood

Independence:

$$\begin{aligned} p(X|\Theta) &= \prod_{n=1}^N p(x_n|\Theta) \\ p(x_n|\Theta) &= \sum_k \pi_k \mathcal{N}(x_n|\mu_k, \Sigma_k) \end{aligned}$$

Thus:

$$\ln p(X|\Theta) = \sum_{n=1}^N \ln \left(\sum_{k=1}^K \pi_k \mathcal{N}(x_n|\mu_k, \Sigma_k) \right)$$

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(b) MLE

$$z_{nk} = \begin{cases} 1 & \text{if } x_n \text{ from cluster } k \\ 0 & \text{otherwise} \end{cases}$$

Complete likelihood:

$$p(X, Z|\Theta) = \prod_n \prod_k [\pi_k \mathcal{N}(x_n|\mu_k, \Sigma_k)]^{z_{nk}}$$

$$\ln p(X, Z|\Theta) = \sum_n \sum_k z_{nk} [\ln \pi_k + \ln \mathcal{N}(x_n|\mu_k, \Sigma_k)]$$

MLE for μ_k

Extract terms:

$$\sum_n z_{nk} (x_n - \mu_k)^T \Sigma_k^{-1} (x_n - \mu_k)$$

Differentiate:

$$\frac{\partial}{\partial \mu_k} \sum_n z_{nk} (x_n - \mu_k)^T \Sigma_k^{-1} (x_n - \mu_k) = -2 \sum_n z_{nk} \Sigma_k^{-1} (x_n - \mu_k)$$

Set to zero:

$$\sum_n z_{nk} (x_n - \mu_k) = 0$$

$$\sum_n z_{nk} x_n = \mu_k \sum_n z_{nk}$$

$$\boxed{\mu_k = \frac{\sum_n z_{nk} x_n}{N_k}}$$

MLE for π_k

$$\boxed{\pi_k = \frac{N_k}{N}}$$

MLE for Σ_k

$$\Sigma_k = \frac{1}{N_k} \sum_n z_{nk} (x_n - \mu_k)(x_n - \mu_k)^T$$